

Infosys - 100 Technical MCQ Questions

Infosys Limited

For BTech Computer Science / IT Students

Object-Oriented Programming

1. Which of the following is true regarding static methods and polymorphism?

- A) Static methods support runtime polymorphism
- B) Static methods are resolved at compile time and do not support runtime polymorphism
- C) Static methods cannot be called
- D) Static methods must be abstract

2. Which concept allows objects of different classes to be treated as objects of a common superclass?

- A) Polymorphism
- B) Encapsulation
- C) Abstraction
- D) Overloading

3. What is a 'friend' function in C++?

- A) A function that has access to private/protected members of a class
- B) A public function only
- C) A function that cannot access class members
- D) A default constructor

4. What is a virtual function used for in C++?

- A) To achieve runtime polymorphism
- B) To avoid inheritance
- C) To create static memory
- D) To improve compilation speed

5. Can a constructor be overridden?

- A) Yes, always
- B) No, constructors cannot be overridden
- C) Only static constructors can
- D) Only in interfaces

6. Which of the following is true about a constructor?

- A) It has a return type
- B) It is called automatically when an object is created
- C) It can be inherited like other methods
- D) It must be private

7. Which OOP principle is primarily used to reduce code duplication by reusing existing classes?

- A) Inheritance
- B) Polymorphism
- C) Abstraction
- D) Encapsulation

8. Which of the following best describes 'abstraction'?

- A) Showing only relevant details and hiding implementation complexity
- B) Combining data and methods
- C) Creating multiple objects
- D) Executing code faster

9. Which of these is true about interfaces in Java 8+?

- A) They can have default and static methods with bodies
- B) They can have instance variables with values changed at runtime
- C) They support constructors
- D) They cannot be implemented by multiple classes

10. A class that cannot be instantiated directly is called a(n):

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

11. In Java, which keyword is used to inherit a class?

- A) implements
- B) extends
- C) inherits
- D) super

12. What is the purpose of the 'super' keyword in Java?

- A) To create a static method
- B) To refer to the parent class members
- C) To delete an object
- D) To define an interface

13. Which of these is an example of compile-time polymorphism?

- A) Method overriding
- B) Operator overloading (in C++)
- C) Dynamic method dispatch
- D) Runtime type identification

14. Constructor overloading is possible when:

- A) Constructors have same name but different parameter lists
- B) Constructors are private
- C) A class has no constructor
- D) Only one constructor exists

15. What happens if a class does not implement all abstract methods of an interface it implements?

- A) Compilation succeeds silently
- B) The class must be declared abstract
- C) Runtime error occurs
- D) Nothing happens

16. What is object composition?

- A) Building complex objects by combining simpler objects
- B) Inheriting from multiple classes
- C) Creating static objects
- D) Deleting objects

17. Which access modifier makes a member accessible only within the same class?

- A) public
- B) protected
- C) private
- D) default

18. What is method hiding in OOP?

- A) When a static method in a subclass has the same signature as one in the superclass
- B) When a private method is called
- C) When a method throws an exception
- D) When constructors are overloaded

DBMS & SQL

19. What is the purpose of an index in a database?

- A) To enforce referential integrity
- B) To speed up data retrieval operations
- C) To store backup data
- D) To encrypt table data

20. What is the purpose of the GROUP BY clause?

- A) To sort data in ascending order
- B) To arrange identical data into groups for aggregate calculations
- C) To filter rows before aggregation
- D) To join multiple tables

21. Which SQL command is used to change the structure of an existing table?

- A) UPDATE
- B) ALTER
- C) MODIFY
- D) CHANGE

22. Which of these correctly describes a many-to-many relationship implementation?

- A) Implemented directly with a foreign key in one table
- B) Implemented using a junction/associative table with foreign keys to both tables
- C) Not possible in relational databases
- D) Requires a trigger

23. What is a composite key?

- A) A key formed by combining two or more columns to uniquely identify a row
- B) A key that references another table
- C) An automatically generated key
- D) A key with only one column

24. Which SQL aggregate function returns the number of rows matching a condition?

- A) SUM()
- B) COUNT()
- C) AVG()
- D) MAX()

25. Which isolation level allows 'dirty reads'?

- A) Serializable
- B) Repeatable Read
- C) Read Committed
- D) Read Uncommitted

26. Which SQL keyword is used to remove duplicate rows from a result set?

- A) UNIQUE
- B) DISTINCT
- C) FILTER
- D) ONLY

27. What is a schema in a database context?

- A) The logical structure that defines tables, columns, and relationships
- B) A backup file
- C) A stored procedure
- D) A query optimizer

28. What is the difference between WHERE and HAVING clauses?

- A) WHERE filters rows before grouping, HAVING filters groups after aggregation
- B) They are identical in function
- C) HAVING is used only with SELECT *
- D) WHERE cannot use comparison operators

29. Which of the following is true about a candidate key?

- A) It can contain NULL values
- B) Every candidate key can be a primary key, but only one is chosen as such
- C) A table can have only one candidate key
- D) It must always be a foreign key

30. Which index type stores data rows in the same order as the index?

- A) Non-clustered index
- B) Clustered index
- C) Bitmap index
- D) Hash index

31. Which SQL statement is used to combine the result-set of two or more SELECT statements with the same number of columns?

- A) JOIN
- B) UNION
- C) MERGE
- D) GROUP BY

32. What is a self-join?

- A) A join of a table with itself
- B) A join between two different databases
- C) A join without any condition
- D) An invalid SQL operation

33. What is a view in SQL?

- A) A physical copy of a table
- B) A virtual table based on the result of a SQL query
- C) A type of index
- D) A stored procedure

34. What is a trigger in SQL?

- A) A special stored procedure automatically executed in response to certain events on a table
- B) A type of index
- C) A join operation
- D) A constraint on primary keys

35. What is denormalization?

- A) The process of adding redundancy to improve read performance
- B) Removing all keys from a table
- C) A type of join
- D) Encrypting a database

36. Which of the following best defines a stored procedure?

- A) A precompiled collection of SQL statements stored in the database and executed as a unit
- B) A type of foreign key
- C) A column constraint
- D) A backup mechanism

37. What does the term 'referential integrity' ensure?

- A) That foreign key values always match a primary key value in the referenced table (or are NULL)
- B) That all columns are indexed
- C) That all queries run faster
- D) That tables cannot be dropped

38. Which normal form eliminates transitive dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 0NF

Operating Systems

39. Which of the following is an example of an inter-process communication (IPC) mechanism?

- A) Pipes
- B) Compilers
- C) Loaders
- D) Linkers

40. What does 'thrashing' refer to in an OS?

- A) Excessive CPU usage by a single process
- B) A state where the system spends most of its time swapping pages instead of executing processes
- C) A type of deadlock
- D) A network congestion issue

41. What is the difference between paging and segmentation?

- A) Paging divides memory into fixed-size blocks; segmentation divides memory into variable-sized logical units
- B) They are identical
- C) Segmentation is faster in all cases
- D) Paging is used only for disk storage

42. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

43. Which of the following best describes multiprogramming?

- A) Running multiple programs on multiple CPUs simultaneously only
- B) Keeping multiple programs in memory so the CPU always has one to execute, improving utilization
- C) Running only one program at a time
- D) A type of scheduling exclusive to real-time systems

44. Which of these describes a 'zombie process'?

- A) A process that has completed execution but still has an entry in the process table
- B) A process that never starts
- C) A process using 100% CPU
- D) A kernel-level thread

45. Which of these is a necessary condition to prevent deadlock by breaking 'circular wait'?

- A) Impose a strict ordering of resource acquisition
- B) Allow unlimited resource requests
- C) Never release resources
- D) Increase the number of processes

46. What is the purpose of a semaphore?

- A) To manage access to shared resources and avoid race conditions
- B) To compile code
- C) To manage file names
- D) To render graphics

47. What is a monolithic kernel?

- A) A kernel where all OS services run in kernel space as a single large process
- B) A kernel with no drivers
- C) A kernel used only in embedded systems
- D) A kernel that cannot be modified

48. Which of the following is a preemptive scheduling algorithm?

- A) FCFS
- B) Round Robin
- C) Non-preemptive SJF
- D) Batch scheduling

49. What is the purpose of a context switch?

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

50. What is a critical section in OS?

- A) A section of code that accesses shared resources and must not be executed by more than one process/thread at a time
- B) A section reserved for the kernel only
- C) A type of page fault
- D) A memory segment for stack only

51. What is spooling?

- A) Simultaneous Peripheral Operations On-Line - buffering data for devices like printers
- B) A type of deadlock
- C) A page replacement algorithm
- D) A CPU scheduling technique

52. Which of the following is true about a microkernel architecture?

- A) It moves most services to user space, keeping the kernel minimal
- B) It has no scheduler
- C) It cannot support IPC
- D) It is always slower with no exceptions

Computer Networks

53. Which of the following best describes latency in networking?

- A) The amount of data transferred per second
- B) The delay before data transfer begins following an instruction
- C) The number of devices on a network
- D) The physical distance between two devices

54. What is NAT (Network Address Translation) used for?

- A) To translate private IP addresses to a public IP address for internet access
- B) To encrypt data
- C) To assign MAC addresses
- D) To resolve domain names

55. What does VPN stand for and what is its main purpose?

- A) Virtual Private Network - creates a secure, encrypted connection over a public network
- B) Virtual Public Network - increases internet speed
- C) Verified Private Node - a type of router
- D) Virtual Protocol Number - an IP addressing scheme

56. Which of the following is a connectionless protocol?

- A) TCP
- B) UDP
- C) FTP
- D) HTTP over TCP

57. What is a firewall used for?

- A) To speed up internet connections
- B) To monitor and control incoming/outgoing network traffic based on security rules
- C) To assign IP addresses
- D) To translate domain names

58. What does 'three-way handshake' refer to in TCP?

- A) The SYN, SYN-ACK, ACK process used to establish a connection
- B) A method to terminate a connection
- C) An error-detection mechanism
- D) A DNS lookup process

59. Which of the following best describes a subnet mask?

- A) It divides an IP address into network and host portions
- B) It encrypts network traffic
- C) It assigns MAC addresses
- D) It is used only in IPv6

60. Which protocol is used to send emails?

- A) POP3
- B) SMTP
- C) HTTP
- D) FTP

61. What is the maximum number of usable host addresses in a /24 subnet?

- A) 256
- B) 254
- C) 255
- D) 512

62. What is the purpose of a subnet?

- A) To divide a large network into smaller, more manageable segments
- B) To increase the number of MAC addresses
- C) To eliminate the need for routers
- D) To replace DNS

63. Which layer of the OSI model handles encryption and data formatting?

- A) Presentation Layer
- B) Session Layer
- C) Application Layer
- D) Physical Layer

64. Which of the following is true about IPv6 compared to IPv4?

- A) IPv6 uses 32-bit addresses while IPv4 uses 128-bit
- B) IPv6 uses 128-bit addresses, providing a vastly larger address space than IPv4's 32-bit addresses
- C) IPv6 is identical to IPv4 in address length
- D) IPv4 has replaced IPv6 entirely

Data Structures & Algorithms

65. Which data structure would you use to check if a string of brackets is balanced?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

66. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

67. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

68. What does BFS stand for in graph traversal?

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

69. What is dynamic programming primarily used for?

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

70. What is the time complexity of Dijkstra's algorithm using a binary heap?

- A) $O(V^2)$
- B) $O(E \log V)$
- C) $O(V + E)$
- D) $O(E^2)$

71. What is the key property of a min-heap?

- A) The parent node is always greater than its children
- B) The parent node is always smaller than or equal to its children
- C) It must be a complete binary tree only, with no ordering
- D) It is always a balanced BST

72. Which algorithm is used to find the Minimum Spanning Tree of a graph?

- A) Dijkstra's Algorithm
- B) Kruskal's or Prim's Algorithm
- C) Bellman-Ford Algorithm
- D) Floyd-Warshall Algorithm

73. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

74. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

75. What is the time complexity of inserting an element at the head of a singly linked list?

- A) $O(n)$
- B) $O(1)$
- C) $O(\log n)$
- D) $O(n \log n)$

76. What is the time complexity of finding the minimum element in an unsorted array?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

- 77. What is the time complexity of Bubble Sort in the worst case?**
- A) $O(n)$
 - B) $O(n \log n)$
 - C) $O(n^2)$
 - D) $O(\log n)$
- 78. Which data structure is typically used to implement BFS?**
- A) Stack
 - B) Queue
 - C) Heap
 - D) Array only
- 79. Which data structure is typically used to implement DFS (iteratively)?**
- A) Queue
 - B) Stack
 - C) Heap
 - D) Hash table
- 80. Which of the following is a common technique to resolve hash collisions?**
- A) Chaining or open addressing
 - B) Binary search
 - C) Recursion
 - D) Memoization
- 81. Which of the following problems is a classic example solved using dynamic programming?**
- A) Binary search
 - B) 0/1 Knapsack problem
 - C) Bubble sort
 - D) Linear search
- 82. Which data structure underlies the implementation of recursion internally?**
- A) Queue
 - B) Stack (call stack)
 - C) Heap (memory heap)
 - D) Graph
- 83. Which algorithmic technique does Merge Sort primarily use?**
- A) Greedy
 - B) Divide and Conquer
 - C) Dynamic Programming
 - D) Backtracking
- 84. What is memoization?**
- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
 - B) A sorting technique
 - C) A type of recursion without a base case
 - D) A method to delete duplicate array elements

Programming Language Fundamentals

85. What is the difference between a shallow copy and a deep copy of a list containing nested lists?

- A) A shallow copy also copies nested objects recursively, deep copy does not
- B) A shallow copy copies references to nested objects, deep copy recursively copies nested objects too
- C) They behave identically
- D) Deep copy is not possible in most languages

86. In Java, what is autoboxing?

- A) Automatic conversion between primitive types and their corresponding wrapper classes
- B) A way to box multiple objects together
- C) A garbage collection technique
- D) A type of casting error

87. What is method chaining?

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

88. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

89. Which of the following best describes garbage collection?

- A) A manual process where the developer frees memory explicitly
- B) An automatic process that reclaims memory occupied by objects no longer in use
- C) A compilation step
- D) A type of exception handling

90. What is the difference between a compiler and an interpreter?

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

91. In C++, what does the 'const' keyword do when applied to a variable?

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

92. Which of the following is true about Python's Global Interpreter Lock (GIL)?

- A) It allows true parallel execution of threads on multiple cores
- B) It ensures only one thread executes Python bytecode at a time in CPython, limiting CPU-bound multithreading performance
- C) It has no impact on multithreading
- D) It is used only in Python 2

93. What is the purpose of a 'static' variable in a class (Java/C++)?

- A) It belongs to the instance and is unique per object
- B) It belongs to the class rather than any instance and is shared across all instances
- C) It can only be used inside loops
- D) It is always private

94. What is duck typing (commonly associated with Python)?

- A) A type system that checks types strictly at compile time
- B) An object's suitability is determined by the presence of certain methods/behavior rather than its explicit type
- C) A type of exception
- D) A memory allocation strategy

Web & Cloud Fundamentals

95. What is the primary difference between HTTP and HTTPS?

- A) HTTPS uses encryption (SSL/TLS) to secure data in transit while HTTP does not
- B) HTTP is faster in all cases
- C) HTTPS cannot use port 443
- D) There is no difference

96. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

97. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

98. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

99. What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

100. What is a RESTful API?

- A) An API that strictly requires SOAP protocol
- B) An architectural style for designing networked applications using stateless operations over standard HTTP methods
- C) A type of database
- D) A programming language

Answer Key

1-B	2-A	3-A	4-A	5-B	6-B	7-A	8-A	9-A	10-A
11-B	12-B	13-B	14-A	15-B	16-A	17-C	18-A	19-B	20-B
21-B	22-B	23-A	24-B	25-D	26-B	27-A	28-A	29-B	30-B
31-B	32-A	33-B	34-A	35-A	36-A	37-A	38-C	39-A	40-B
41-A	42-B	43-B	44-A	45-A	46-A	47-A	48-B	49-A	50-A
51-A	52-A	53-B	54-A	55-A	56-B	57-B	58-A	59-A	60-B
61-B	62-A	63-A	64-B	65-B	66-C	67-A	68-B	69-A	70-B
71-B	72-B	73-B	74-C	75-B	76-C	77-C	78-B	79-B	80-A
81-B	82-B	83-B	84-A	85-B	86-A	87-A	88-B	89-B	90-A
91-B	92-B	93-B	94-B	95-A	96-A	97-A	98-B	99-A	100-B

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.